

A2 Level Computer Science (9618) — Key Facts & Reference

Cambridge International A2 9618 (Paper 3 & 4). AS knowledge assumed — this sheet covers the A2-only material.

DATA TYPES & FILE ORGANISATION

User-defined types	TYPE name = declared type; enumerated: TYPE day = (mon,tue,...); pointer: ^type
Records & sets	RECORD fields ENDRECORD; SET OF values (membership, union)
File organisation	serial (arrival order), sequential (key order), random/direct (hash to address)
Hashing	address = hash(key) e.g. key MOD tableSize — enables O(1) random access

ADVANCED DATA STRUCTURES

Stack (LIFO)	push, pop, peek — used in recursion, undo
Queue (FIFO)	enqueue, dequeue — used in schedulers, buffers; circular queue wraps
Linked list	node = data + pointer; insert/delete O(1) once located; search O(n)
Binary tree	each node "d 2 children; insertion/search O(log n) if b
Tree traversal	in-order (left,root,right ! sorted), pre-order (root first ! delete)
Hash table	key ! hash ! bucket; collision ! chaining or open address
Graphs	adjacency list/matrix; BFS (queue, shortest unweighted), DFS (stack)

ALGORITHMS (know these cold for Paper 4)

Insertion sort	build sorted section by shifting — O(n ²), good for nearly-sorted
Bubble sort	adjacent swaps until no swaps — O(n ²)
Merge/quick sort	divide & conquer — O(n log n)
Binary search	halve sorted data each step — O(log n)
Dijkstra's	shortest path, greedy, priority queue — non-negative weights
Big-O	$O(1) < O(\log n) < O(n) < O(n \log n) < O(n^2) < O(2^n)$ — time

RECURSION & OOP

Recursion	base case + self-call; each call gets its own stack frame
OOP pillars	encapsulation (private data + methods), inheritance (subclass extends), polymorphism (override)
Class syntax	CLASS name; PRIVATE/PUBLIC; CONSTRUCTOR; NEW instance; THIS/self
Containment	has-a (object as attribute) vs is-a (inheritance)

SYSTEMS & REGISTERS (Paper 3)

CISC vs RISC	RISC: simple fixed instructions, pipelining, more registers; CISC: complex multi-cycle
Pipelining	overlap FDE stages; hazard = flush or stall
Interrupt handling	ISR saved state, serviced, restored — priorities matter
Parallel systems	SISD/SIMD/MISD/MIMD; Amdahl: speedup limited by serial fraction
Memory hierarchy	registers ! cache ! RAM ! disk (speed !, size !, cost/bi
AS-level numbers	sign & magnitude, two's complement, BCD (per-digit 4 bits), floating point normalisation

DATABASES & SQL (Paper 3)

DDL	CREATE DATABASE/TABLE; ALTER; DROP
DML	SELECT cols FROM t WHERE ... ; INSERT INTO; UPDATE ... SET; DELETE FROM
Joins & grouping	JOIN t2 ON key; GROUP BY + COUNT/SUM/AVG; ORDER BY; DISTINCT
Normalisation	1NF atomic values; 2NF no partial dependency; 3NF no transitive dependency

SIMULATION & PROBLEM SOLVING (Paper 4)

Simulation loop	read time/state !' update model !' output; use real-time
Algorithm design	pseudocode first, then trace table, then code — marks for every stage